

Week 5: Simple Linear Regression I

PLSC 309 Quantitative Political Analysis

Yang Yang

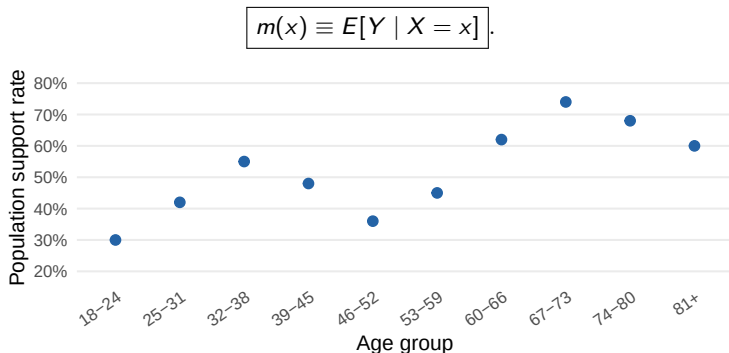
Department of Political Science
Penn State

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- Conditional Expectation Function (CEF)
- Best Linear Predictor (BLP)
- Ordinary Least Squares (OLS)

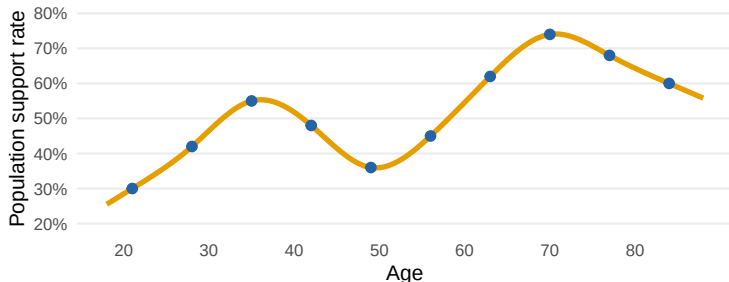
Review: Conditional Expectation Function

- Broadly speaking we are interested in the conditional expectation function (CEF).
- It tells the average value of one variable given specific values of another variable.
- As you change the value of x , the expected (average) value of Y changes too.



Each point represents the population support rate within an age group.

Review: Conditional Expectation Function



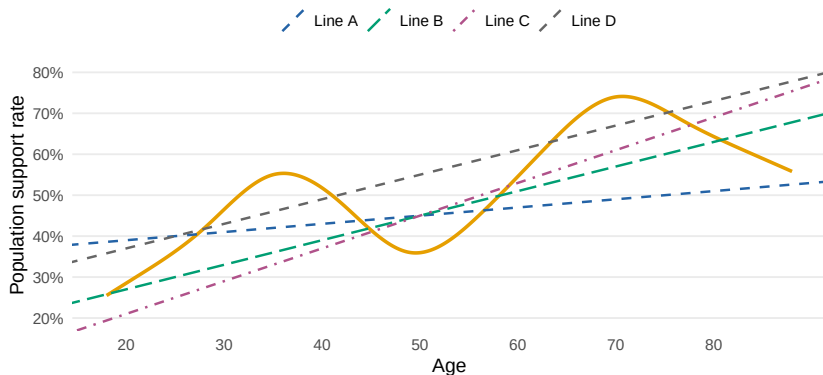
When age is treated as a continuous variable, we can represent the population support rate as a smooth function of age. The smooth curve shows the population support rate at each age.

- The CEF might have a potentially arbitrary shape and can be infinitely complex.
- Estimating the CEF can therefore be rather difficult.
- So, instead we can approximate the CEF with linear predictors.
- A best linear predictor!

Review: Linear Predictors

$$g(X) = a + bX.$$

- a is the intercept and b is the slope.



Which line should we choose, and what makes it the best?

Review: Best Linear Predictor

Definition

The **Best Linear Predictor (BLP)** of Y given X is the linear function $\alpha + \beta X$ that minimizes the population mean squared prediction error:

$$(\alpha, \beta) = \arg \min_{a, b} E[(Y - a - bX)^2].$$

For any candidate line $a + bX$:

- $Y - (a + bX)$ is the prediction error.
- Squaring the error penalizes larger mistakes more heavily.
- Taking the expectation averages these squared errors over the population.

“Best” means **the smallest mean squared prediction error among all linear predictors**.

Review: How Do We Find the BLP?

Choose the intercept and slope to minimize population mean squared prediction error:

$$Q(a, b) = E[(Y - a - bX)^2], \quad (\alpha, \beta) = \arg \min_{a, b} Q(a, b).$$

1. Differentiate

With respect to a :

$$\begin{aligned} \frac{\partial Q}{\partial a} &= E[2(Y - a - bX)(-1)] \\ &= -2E[Y - a - bX]. \end{aligned}$$

With respect to b :

$$\begin{aligned} \frac{\partial Q}{\partial b} &= E[2(Y - a - bX)(-X)] \\ &= -2E[X(Y - a - bX)]. \end{aligned}$$

Set both derivatives to zero at (α, β) :

$$E[Y - \alpha - \beta X] = 0, \quad (1)$$

$$E[X(Y - \alpha - \beta X)] = 0. \quad (2)$$

2. Solve for the intercept

Expand condition (1):

$$E[Y] - \alpha - \beta E[X] = 0.$$

Rearrange:

$$\boxed{\alpha = E[Y] - \beta E[X].}$$

Next, expand condition (2):

$$E[XY] - \alpha E[X] - \beta E[X^2] = 0.$$

Substitute $\alpha = E[Y] - \beta E[X]$:

$$\begin{aligned} 0 &= E[XY] \\ &\quad - (E[Y] - \beta E[X])E[X] \\ &\quad - \beta E[X^2]. \end{aligned}$$

3. Substitute and solve for the slope

Expand and collect terms:

$$\begin{aligned} 0 &= E[XY] - E[X]E[Y] \\ &\quad + \beta(E[X])^2 - \beta E[X^2] \\ &= \underbrace{E[XY] - E[X]E[Y]}_{\text{Cov}(X, Y)} \\ &\quad - \beta \underbrace{(E[X^2] - (E[X])^2)}_{\text{Var}(X)}. \end{aligned}$$

Therefore,

$$\beta \text{Var}(X) = \text{Cov}(X, Y).$$

Divide by $\text{Var}(X) > 0$:

$$\boxed{\beta = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}}.$$

The result: $\text{BLP}(X) = \alpha + \beta X$

Intercept: $\alpha = E[Y] - \beta E[X]$

The line's predicted value at $X = 0$.

It makes the line pass through $(E[X], E[Y])$.

Slope: $\beta = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$

A one-unit increase in X changes the linear prediction by β units.

Theorem

The **Best Linear Predictor** of Y given X is

$$g(X) = \alpha + \beta X,$$

where

$$\alpha = E[Y] - \beta E[X], \quad \beta = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}.$$

- This may not be a good approximation depending on how non-linear the true CEF is. However, it provides us with a reasonable target that always exists.

Review: What is OLS?

The **BLP** chooses the line that minimizes population MSE:

$$(\alpha, \beta) = \arg \min_{a, b} E[(Y - a - bX)^2].$$

In practice, we observe a sample rather than the entire population. We replace the population expectation with a sample average:

$$E[(Y - a - bX)^2] \longrightarrow \frac{1}{n} \sum_{i=1}^n (Y_i - a - bX_i)^2.$$

Ordinary Least Squares (OLS)

OLS estimates the BLP coefficients by **minimizing the sum of the squared residuals** (sample analog of the mean squared predication error):

$$(\hat{\alpha}, \hat{\beta}) = \arg \min_{a, b} \sum_{i=1}^n (Y_i - a - bX_i)^2.$$

Review: BLP vs. OLS

Population BLP

$$\beta = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

$$\alpha = E[Y] - \beta E[X]$$

Sample OLS

$$\hat{\beta} = \frac{\widehat{\text{Cov}}(X, Y)}{\widehat{\text{Var}}(X)}$$

$$\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{X}$$

OLS Slope and Intercept

$$\hat{\beta} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}, \quad \hat{\alpha} = \bar{Y} - \hat{\beta}\bar{X}.$$

- Numerator: sample covariation of X and Y
- Denominator: sample variation of X
- $\hat{\beta} = 0$ iff X and Y are sample-uncorrelated

Hand-Computing $\hat{\beta}$ for the Six Fair Elections

X= GDP Growth Rate, Y=Incumbent Vote Share

Year	X_i	Y_i	$X_i - \bar{X}$	$Y_i - \bar{Y}$	$(X_i - \bar{X})(Y_i - \bar{Y})$	$(X_i - \bar{X})^2$
1956	3.1	57.8	-0.75	+2.92	-2.19	0.56
1960	0.1	49.9	-3.75	-4.98	+18.68	14.06
1964	5.3	61.3	+1.45	+6.42	+9.31	2.10
1968	5.1	49.6	+1.25	-5.28	-6.60	1.56
1972	5.9	61.8	+2.05	+6.92	+14.18	4.20
1976	3.6	48.9	-0.25	-5.98	+1.50	0.06
Sum	23.1	329.3	0	0	34.88	22.55
Mean	3.85	54.88				

OLS Coefficients

$$\hat{\beta} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{34.88}{22.55} \approx 1.55$$

$$\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{X} = 54.88 - (1.55)(3.85) \approx 48.9$$

Verifying with R

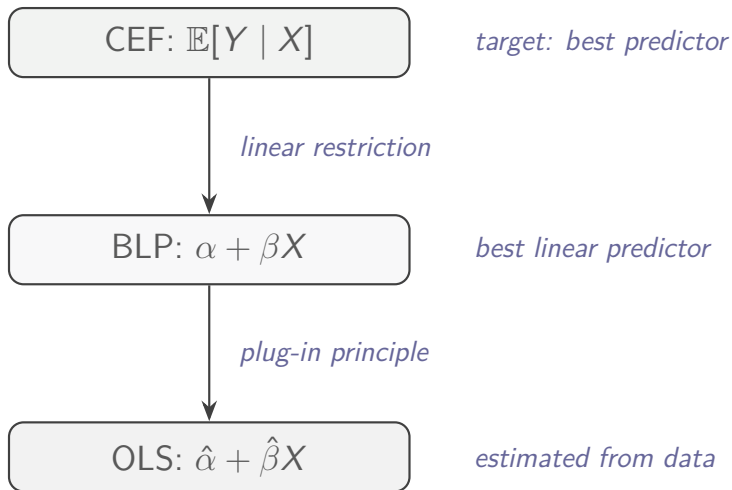
```
# Enter the data
fair <- data.frame(
  Year = c(1956, 1960, 1964, 1968, 1972, 1976),
  X = c(3.1, 0.1, 5.3, 5.1, 5.9, 3.6),
  Y = c(57.8, 49.9, 61.3, 49.6, 61.8, 48.9)
)
```

```
# Estimate the OLS regression
model <- lm(Y ~ X, data = fair)
```

```
# Regression coefficients
coef(model)
```

```
## (Intercept)          X
##      48.93038      1.54622
```

Review: CEF, BLP, and OLS



Today's Agenda

- Fitted values
- Residuals
- Interpreting regression coefficients
- How measurement units affect coefficients
- Estimating a regression in R
- Review of Fundamentals of Probability and Statistics

Fitted Values and Residuals

Definition (Fitted Value)

A **fitted value** or **predicted value** is the value of Y_i predicted by the estimated regression line for a given value of X_i :

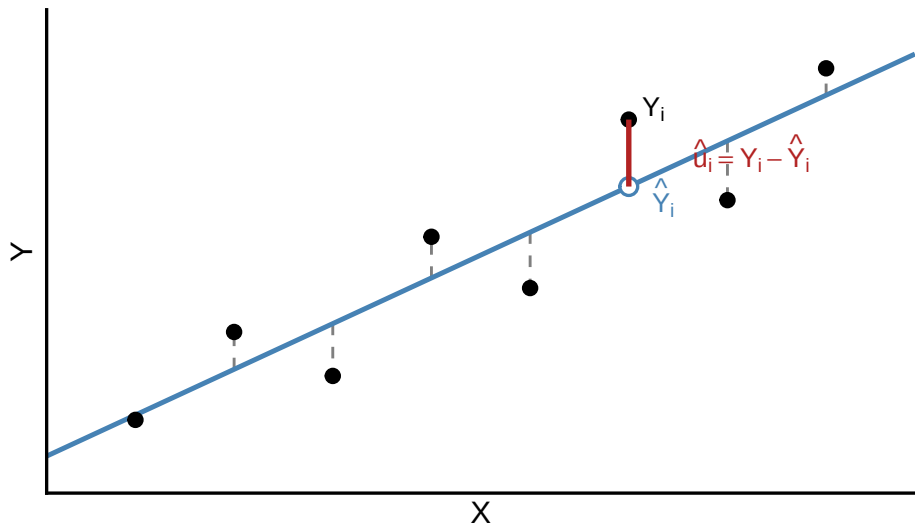
$$\hat{Y}_i = \hat{\alpha} + \hat{\beta}X_i$$

Definition (Residual)

The **residual** is the difference between the observed value of Y_i and its predicted value:

$$\hat{u}_i = Y_i - \hat{Y}_i = Y_i - \hat{\alpha} - \hat{\beta}X_i$$

Fitted Values and Residuals



Population: BLP

$$Y_i = \underbrace{\alpha + \beta X_i}_{\text{Population prediction}} + u_i$$

- Y_i : outcome for individual i .
- X_i : predictor for individual i .
- α : population intercept.
- β : population slope.
- u_i : prediction error relative to the BLP.

Sample: OLS

$$Y_i = \underbrace{\hat{\alpha} + \hat{\beta} X_i}_{\text{Fitted value } \hat{Y}_i} + \hat{u}_i$$

- Y_i : the same observed outcome.
- X_i : the same observed predictor.
- $\hat{\alpha}$: estimated intercept.
- $\hat{\beta}$: estimated slope.
- $\hat{u}_i$: residual relative to the fitted line.

OLS uses sample data to estimate the population BLP coefficients.

Interpreting the Slope Coefficient $\hat{\beta}$

Recall the estimated regression line:

$$\hat{Y}_i = \hat{\alpha} + \hat{\beta}X_i$$

Interpretation of $\hat{\beta}$

A **one-unit increase in X** is associated with an average change of $\hat{\beta}$ **units in the predicted value of Y** .

Why?

$$\begin{aligned}\hat{Y}(X + 1) &= \hat{\alpha} + \hat{\beta}(X + 1) \\ &= \hat{\alpha} + \hat{\beta}X + \hat{\beta}\end{aligned}$$

Therefore,

$$\hat{Y}(X + 1) - \hat{Y}(X) = \boxed{\hat{\beta}}$$

Interpreting $\hat{\beta}$: Sign and Magnitude

The coefficient $\hat{\beta}$ tells us both the **direction** and **magnitude** of the relationship.

- $\hat{\beta} > 0$:
Higher values of X are associated with higher predicted values of Y .
- $\hat{\beta} < 0$:
Higher values of X are associated with lower predicted values of Y .
- $\hat{\beta} = 0$:
There is no linear relationship between X and the predicted value of Y .

Magnitude

The absolute size of $\hat{\beta}$ tells us how much predicted Y changes when X increases by one unit.

Intuition of the OLS Estimator

- The intercept equation tells us that the regression line goes through the point $(\bar{X}, \bar{Y})$:

$$\bar{Y} = \hat{\alpha} + \hat{\beta}\bar{X}$$

- The slope of the regression line can be written as:

$$\hat{\beta} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

- The larger the **covariance** between X and Y , the larger the **slope**, holding the variance of X fixed.
- Negative covariance $\Rightarrow$ negative slope;
positive covariance $\Rightarrow$ positive slope.
- If X_i does not vary, $\text{Var}(X) = 0$, so the slope is undefined.
- If Y_i does not vary, $\text{Cov}(X, Y) = 0$, so the estimated slope is zero.

Example: Interpreting $\hat{\beta}$

Suppose we estimated:

$$\hat{Y} = 48.93 + 1.55X$$

where

- X is an explanatory variable,
- Y is the outcome measured in percentage points.

Since

$$\hat{\beta} = 1.55,$$

a **one-unit increase in X** is associated with an average

1.55-point increase in predicted Y .

For example:

$$\begin{aligned} X = 3 &\Rightarrow \hat{Y} = 48.93 + 1.55(3) = 53.58 \\ X = 4 &\Rightarrow \hat{Y} = 48.93 + 1.55(4) = 55.13 \end{aligned}$$

Therefore,

$$55.13 - 53.58 = 1.55.$$

Always Pay Attention to Units

The interpretation of $\hat{\beta}$ depends on the units of both X and Y .

Suppose:

$$\hat{\beta} = 2.5$$

Example 1

If X is measured in **years** and Y is measured in **points**:

$$+1 \text{ year in } X \Rightarrow +2.5 \text{ points in predicted } Y.$$

Example 2

If X is measured in **thousands of dollars**:

$$+\$1,000 \text{ in } X \Rightarrow +2.5 \text{ units in predicted } Y.$$

Always interpret coefficients using the original units of X and Y .

Units Matter: Changing the Units of Y

Suppose X is education in years and Y is annual income.

```
data <- data.frame(
  education = c(12, 14, 16, 18, 20),
  income_dollars = c(40000, 50000, 60000, 70000, 80000)
)

# Y measured in dollars
m1 <- lm(income_dollars ~ education, data = data)
coef(m1)
```

```
## (Intercept)    education
##          -20000         5000
```

Now measure income in **thousands of dollars**:

```
data$income_thousands <- data$income_dollars / 1000
m2 <- lm(income_thousands ~ education, data = data)
coef(m2)
```

```
## (Intercept)    education
##          -20         5
```

How Units Affect Regression Coefficients

Key Point

The numerical size of a regression coefficient depends on the **units of measurement**.

A “large” coefficient does not necessarily mean a strong relationship.

Review: Fundamentals of Probability and Statistics